

SyntheticCAD

Synthetic Law Enforcement CAD Data Generator

Product Requirements Document — MVP

Coding It Forward • Hack Your Summer • Session A

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Point of Contact: David Schwindt

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1. Purpose and How to Use This Document

This document is your project charter, technical specification, and onboarding guide. If you are joining this project as an intern, start here and read it fully before writing a single line of code. It explains what you are building, who will use it, why it matters, and what success looks like.

You do not need prior experience with law enforcement data, dispatch systems, or synthetic data generation to contribute. Everything you need to understand the domain is explained in this document.

This is a real project with real potential users. Your output at the end of four weeks should be something that a non-technical law enforcement agency could install, run, and use to enable academic research partnerships.

2. Background: The Problem This Project Solves

2.1 The Research Gap in Law Enforcement Data

Academic researchers who want to study law enforcement operations face a significant barrier: agencies are reluctant or unable to share their operational data. The reasons are legitimate. Law enforcement data contains personally identifiable information (PII) about crime victims, witnesses, and individuals who were investigated but never charged. It is subject to strict federal standards called CJIS (Criminal Justice Information Services) requirements, which impose storage, access, and handling obligations that most universities cannot easily meet. Sharing this data exposes agencies to legal liability.

The result is a research gap. Patterns in public safety operations, response times, call volumes, and resource deployment go unstudied because the underlying data is locked behind legal and compliance barriers that researchers cannot practically clear.

2.2 The Solution: Synthetic Data

Synthetic data is artificially generated data that mirrors the statistical patterns of a real dataset while containing no real records, no real people, and no identifiable information. Think of it as a statistically faithful copy that has no connection to any individual incident or individual.

A synthetic law enforcement dataset would allow researchers to:

- Study call volume patterns and response time distributions
- Test analytical models and policy hypotheses
- Publish findings without accessing or storing real PII
- Reproduce and audit each other's methodology

Critically, synthetic data has no value to someone trying to re-identify specific individuals or incidents because those individuals and incidents do not exist in it.

2.3 What SyntheticCAD Does

SyntheticCAD is an offline Windows desktop application. A law enforcement agency runs it on their own computer. The application ingests the agency's real operational data, learns its statistical patterns, and generates a completely synthetic dataset that preserves those patterns. The synthetic dataset can then be released to researchers with no CJIS obligations and no PII exposure.

The application never sends data anywhere. Everything runs locally on the agency's machine.

3. Understanding CAD Data

CAD stands for Computer-Aided Dispatch. CAD data is the operational record generated by a 911 dispatch center every time someone calls for police, fire, or emergency services. Understanding the structure of this data is essential before you write any code.

3.1 What CAD Data Records

Each entry in a CAD dataset represents a call for service. A call for service begins when someone contacts dispatch (by phone, radio, or officer self-initiation) and ends when the responding officers clear the scene. A typical record captures:

Field	What it captures
Event ID	A unique identifier for each call for service
Date and Time	When the call came in
Location	Address or geographic coordinates of the incident
Call Type	Category of the call (e.g., traffic stop, burglary, welfare check)
Priority	How urgently the call needs a response (usually 1-4 or 1-5)
Disposition	How the call was resolved (e.g., report taken, arrest made, unfounded, no service needed)
Unit ID	Identifier for the officer or unit assigned to the call
Dispatch Time	When the officer was assigned to the call
Arrival Time	When the officer arrived on scene
Clearance Time	When the officer cleared the scene and became available again

3.2 The One-to-Many Relationship: Events and Units

This is the most important structural concept in CAD data. Read it carefully.

A single call for service can involve multiple officers. A robbery call might dispatch three officers. Each officer has their own dispatch time, arrival time, and clearance time. In a CAD export, this means one event (the robbery) appears as multiple rows in the CSV file, one row per officer assigned.

Example

Event #12345 is a robbery at 3rd and Main. Officer A is dispatched at 10:02, arrives at 10:07, clears at 11:15. Officer B is dispatched at 10:02, arrives at 10:09, clears at 10:45. Officer C is dispatched at 10:14 (called for backup), arrives at 10:19, clears at 11:15. In the CSV export, this single robbery appears as three rows, all sharing Event ID 12345 but with different unit IDs and timestamps.

When your application ingests CAD data, it must recognize this structure. The statistical patterns in the event-level data (call volumes, call types, locations) are different from the patterns in the unit-level data (response times, how many units respond to each call type). Your synthetic data generation must handle both levels appropriately.

During the field mapping step of the application, you will guide the user through identifying which fields belong to the event level and which belong to the unit level. This mapping is what makes the synthesis accurate.

3.3 Why CAD Data for This MVP

CAD data is the right starting point for this project for several reasons. In many states it is considered public record, which reduces the legal complexity of working with it. It is less sensitive than arrest records or investigative case files, which reduces privacy risk. It has clear, consistent statistical patterns (call volumes peak at certain hours, response times follow predictable distributions, calls cluster geographically) that synthetic data generation can learn and replicate reliably. And it can be exported from most dispatch platforms as a standard CSV file.

3.4 Development Dataset

You do not need to source real agency data to begin development. The Police Data Initiative (policedatainitiative.org) publishes calls for service data from more than 120 law enforcement agencies as publicly available CSV downloads. Agencies such as Baltimore, Cincinnati, and Bloomington, Indiana have published this data openly.

You should download one or more of these datasets at the start of the project and use them for development and testing. They have real statistical patterns and real field structures. Using them means you are building against actual data rather than a contrived test fixture.

4. Who You Are Building For

Every design and technical decision you make should be made with one of these users in mind. When you are debating between two approaches, ask yourself: which one better serves the person who will actually use this output?

4.1 Law Enforcement Executive (Chief of Police / Sheriff)

This person authorizes the use of the tool and decides whether to release the resulting synthetic dataset to researchers. They are not technical. They need to be able to explain what the tool does to a city council member, a legal team, or a skeptical reporter. Their core fears are liability, embarrassment, and losing control of how their agency is portrayed in research.

	Detail
Goals	Use data to benefit the community. Enable research partnerships. Maintain control over attribution.
Fears	Being named in critical research. Releasing data that could embarrass the agency. Legal liability.
Needs from SyntheticCAD	A plain-language validation summary they can share publicly. An executive dashboard showing real vs. synthetic comparison in visual, non-technical terms.

4.2 Academic Researcher

This person receives the synthetic dataset and uses it to conduct research. They understand statistics. They need to assess whether the synthetic data is of sufficient quality to support peer-reviewed

publication. Their core fear is publishing findings based on data that does not accurately represent reality.

	Detail
Goals	Access realistic law enforcement data without CJIS compliance barriers.
Fears	Publishing flawed research. Being unable to validate data quality to peer review standards.
Needs from SyntheticCAD	A statistical validation report with distribution matching metrics, correlation preservation analysis, and methodology documentation.

4.3 University IRB Reviewer

IRB stands for Institutional Review Board. This is the ethics committee at a university that must approve research involving human subjects data. They need to confirm that the synthetic dataset does not contain real individuals and cannot be reverse-engineered to identify them.

	Detail
Goals	Approve research involving the synthetic dataset under human subjects research standards.
Fears	Synthetic data that could be reverse-engineered to identify real individuals.
Needs from SyntheticCAD	Technical documentation of the anonymization methodology. A clear statement that no real individuals or incidents are represented. Boilerplate disclaimer language that travels with the dataset.

4.4 Privacy and Victim Advocate

This persona represents victims of crime, witnesses, and individuals who interacted with law enforcement but were never charged. Even though synthetic data does not contain real records, advocates want technical assurance that real individuals cannot be inferred from the synthetic output.

	Detail
Goals	Ensure no real person can be identified through the synthetic data, directly or through linkage attacks.
Fears	Sensitive crime types (sexual assault, domestic violence, juvenile incidents) being represented in released data.
Needs from SyntheticCAD	Clear disclaimer language. Documentation of what data was excluded or anonymized.

4.5 Public Policy Analyst

This researcher uses the synthetic data to evaluate the impact of operational decisions on community outcomes. They need data that accurately preserves temporal and geographic patterns.

	Detail
Goals	Evaluate staffing levels, deployment patterns, and response time targets. Identify policy improvements.
Fears	Drawing incorrect policy conclusions from data that does not reflect real-world patterns.
Needs from SyntheticCAD	Data that preserves temporal and geographic distributions. Clear fidelity metrics.

4.6 Agency IT Staff

This is the person at the law enforcement agency who actually installs and runs the application. They may have general IT skills but no Python, data science, or command line experience. If the tool requires them to open a terminal, it has already failed.

	Detail
Goals	Install and run the tool without needing outside help or specialized expertise.
Fears	The tool breaking without support. Not understanding what the tool is doing to their data.
Needs from SyntheticCAD	A standalone Windows .exe. Step-by-step guided workflow. Readable logs. No Python required.

5. Project Scope

5.1 What You Are Building (In Scope)

- Ingest a CAD CSV export from a law enforcement agency
- Guide the user through mapping their CSV fields to the expected schema
- Handle the one-to-many event/unit data structure during ingestion
- Generate synthetic CAD data using an open-source library
- Produce an executive validation dashboard (visual, non-technical)
- Produce a researcher validation report (statistical metrics)
- Export the synthetic dataset as a CSV with accompanying disclaimer language
- Package the application as a standalone Windows desktop executable

5.2 What You Are Not Building (Phase 2)

The following features are explicitly out of scope for this MVP. Do not spend time on them. They are documented here so you understand where the project is headed, not as an invitation to build them now.

- Re-identification and linkage attack testing
- Differential privacy implementation

- Crime type filtering and exclusion controls
 - Audit logging and access tracking
 - Data usage agreement generation
 - Support for data types other than CAD (arrests, incidents, persons)
 - Reproducibility versioning and change logs
 - Web application or cloud deployment
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6. Feature Requirements

6.1 Data Ingestion and Field Mapping

The application must accept a CSV file upload and walk the user through mapping their columns to the standard schema. Not all agencies use the same field names. One agency might call it "IncidentID" and another might call it "EventNumber." The mapping step bridges that gap.

Fields the user must be able to map:

- Event ID (unique identifier for each call for service)
- Call received date and time
- Location (address or latitude/longitude)
- Call type or incident type
- Priority level
- Unit ID (officer or unit identifier)
- Dispatch time (when unit was assigned)
- Arrival time (when unit arrived on scene)
- Clearance time (when unit cleared the scene)
- Disposition (how the call was resolved)

The application must also prompt the user to identify which fields are at the event level (shared across all units on a call) versus the unit level (unique per unit). This is the event/unit grain distinction described in Section 3.

Design Note

The field mapping step is the first thing the user encounters after uploading their file. It sets the tone for the entire experience. If it is confusing, the law enforcement IT persona (Section 4.6) will stop here. Invest in making this step clear, with plain-language labels and examples of what each field means in operational terms.

6.2 Synthetic Data Generation

Once the data is ingested and mapped, the application generates a synthetic dataset that statistically mirrors the original.

The recommended starting point is SDV (Synthetic Data Vault), an open-source Python library that handles relational data structures, which is important for the event/unit relationship in CAD data. The team may evaluate and choose alternatives, but any alternative must handle multi-table relational synthesis.

Alternatives worth evaluating:

- SynthCity (<https://github.com/vanderschaarlab/synthcity>)
- MOSTLY AI open SDK

Whatever library you choose, document your decision and your reasoning. The researcher persona needs to understand the methodology.

The generation process must run entirely offline. No data should leave the user's machine at any point.

6.3 Executive Validation Dashboard

After generation, the application produces a visual dashboard comparing the real data to the synthetic data. This output is designed for the law enforcement executive, not for a statistician. Use plain language. Avoid jargon. The goal is for a Chief of Police to look at this dashboard and feel confident they can share it with their city council or legal team.

Required visualizations:

- Call volume heatmap by day of week and hour of day (real vs. synthetic side by side)
- Response time distribution showing dispatch-to-arrival times (real vs. synthetic)
- Geographic distribution of call locations (pattern-based, not address-level)
- Plain-language summary paragraph describing how closely the synthetic data matches the original

6.4 Researcher Validation Report

The researcher validation report is a separate, more technical output intended for academic audiences and IRB review. It documents the statistical quality of the synthetic data in terms that a peer reviewer or ethics board can evaluate.

Required content:

- Kolmogorov-Smirnov (KS) test results for key numeric fields (see Section 7 for context)
- Correlation preservation analysis between related fields
- Documentation of the synthesis methodology and library used
- Clear statement of data limitations and edge cases
- Statement that the dataset contains no real individuals or real incidents

Note for the Statistics Intern

The metrics listed above are a starting point, not a final prescription. You are expected to review the academic literature on synthetic tabular data validation, evaluate whether these metrics are appropriate, and propose any additions or modifications with documented reasoning. Discuss proposed changes with David Schwindt before implementation. Your methodology section in the validation report should explain why you chose the metrics you did.

6.5 Export and Disclaimer

The synthetic dataset is exported as a CSV file. Every export must include the following disclaimer, either embedded as a header row in the CSV or as an accompanying text file:

Required Disclaimer Language

Research findings derived from this dataset should disclose that the analysis was conducted on synthetic data, not real-world data. Significant effort was made to preserve statistical accuracy, but findings are not guaranteed and should be validated against real data before operational use. The source law enforcement agency may not be named in any published findings without explicit, written permission from that agency.

This disclaimer language is not optional and must not be modified without authorization from the project point of contact.

6.6 Application Packaging

The application must be packaged as a standalone Windows executable (.exe) that does not require the end user to install Python, manage dependencies, or use a command line. The recommended tool is PyInstaller, which bundles a Python application and all its dependencies into a single executable.

The install experience should be: download the .exe, double-click it, and the application opens. That is the standard the agency IT persona (Section 4.6) expects.

The application must produce readable log output so that a non-technical user can see what steps ran, confirm that processing completed successfully, and share the log with support if something goes wrong.

7. Success Metrics

7.1 Quantitative Targets

The following thresholds define the minimum acceptable fidelity for the synthetic dataset. These are starting targets. The statistics intern should evaluate these, research the literature, and propose modifications with documented reasoning.

Metric	Target	Method
Call volume by day and hour	Within 10% of original	Comparative counts
Response time distribution	Within 15% of original	Distribution comparison
KS test on key numeric fields	Document p-values and interpret	Kolmogorov-Smirnov test
Correlation preservation	Preserve sign and approximate magnitude of correlations between related fields	Correlation matrix comparison
Geographic clustering	Visually similar distribution	Qualitative / map overlay

7.2 Research Context

Academic literature on synthetic tabular data validation uses multiple complementary metrics rather than a single threshold. One well-regarded approach is Train on Synthetic, Test on Real (TSTR): train a classifier on the synthetic dataset, test it on the real dataset, and compare performance to a classifier

trained and tested on real data only (TRTR). If the TSTR classifier performs close to the TRTR classifier, that is strong evidence that the synthetic data preserves the predictive structure of the real data. This is worth exploring as a stretch goal if time permits.

7.3 Qualitative Targets

- A non-technical law enforcement executive can review the validation dashboard and understand it without explanation
- An academic researcher can read the validation report and assess whether the data is suitable for their purposes
- The application installs and runs on a Windows machine without errors or command line interaction

8. Data Governance and Researcher Obligations

When a law enforcement agency uses SyntheticCAD to generate and release a synthetic dataset, researchers who use that dataset take on specific obligations. These obligations are communicated through the disclaimer language that travels with every export (see Section 6.5) and are designed to protect both the agency and the integrity of published research.

8.1 Researcher Obligations

- Disclose in any published findings that the analysis was conducted on synthetic data, not real-world data
- Represent that significant effort was made to preserve statistical accuracy, supported by the tool's validation report
- Not name the source agency in any published findings without explicit, written permission from that agency

8.2 Agency Attribution

An agency that releases a synthetic dataset retains the right to decide whether they are named in research based on that data. The default is that they are not named. If a researcher wishes to attribute the data to a specific agency, they must obtain explicit written permission from that agency. This protects agencies from being associated with research conclusions they may disagree with or have had no opportunity to review.

This should be framed to agencies as empowerment, not restriction. They control how their data is used in published work.

9. Technical Guidance

The following are recommendations based on what is most likely to work well in a four-week sprint. They are not mandates. The team should evaluate options together, make decisions as a group, and document their reasoning. If you choose a different tool, be prepared to explain why.

Component	Recommendation	Why
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Language	Python	Strong library ecosystem for data science and packaging
Synthetic generation	SDV (Synthetic Data Vault)	Handles relational data, well-documented, active community
UI framework	Streamlit or Gradio	Minimal code to produce a functional desktop-style interface in Python
Visualization	Plotly	Interactive charts, works well with Streamlit
Packaging	PyInstaller	Bundles Python apps into a standalone Windows .exe
Version control	Team's choice	GitHub is recommended; whatever the team uses, use it consistently

Critical Requirement

The application must run fully offline. No data should leave the user's machine at any point during ingestion, generation, or export. This is a non-negotiable requirement for any law enforcement agency to consider using this tool.

9.1 Open Source References

- SDV (Synthetic Data Vault): <https://github.com/sdv-dev/SDV>
- SynthCity: <https://github.com/vanderschaarlab/synthcity>
- Gretel Synthetics (archived): <https://github.com/gretelai/gretel-synthetics>
- Awesome Synthetic Data (curated tool list): <https://github.com/statice/awesome-synthetic-data>

10. Recommended Team Composition and Roles

This project is designed for a team of three to four people. Each person should have a clearly defined primary responsibility and a clear output they own. The roles below reflect the skill mix most likely to succeed in a four-week sprint.

10.1 Software Developer (1-2 people)

- Primary owner of the data ingestion pipeline, synthetic generation engine, UI, and application packaging
- Python development experience required
- Comfortable installing and working with Python libraries and build tools

10.2 Statistics or Data Science Student (1 person)

- Primary owner of the validation methodology and researcher validation report
- Responsible for evaluating the prescribed metrics in Section 7, reviewing the literature, and proposing any modifications with documented reasoning
- Background in statistical analysis, distributions, and data quality assessment
- This role is critical for satisfying the researcher and IRB reviewer personas

10.3 Public Policy or Criminal Justice Student (1 person)

- Primary owner of the executive dashboard content and plain-language outputs
- Responsible for ensuring the tool's outputs are meaningful and interpretable to law enforcement practitioners and policymakers
- Helps the team ground technical decisions in real-world operational context
- Does not need to write code, but should be involved in design decisions that affect the user-facing outputs

11. Suggested Sprint Plan

Session A runs June 15 through July 10: four weeks. The following is a suggested structure. The team should adapt it based on their actual pace and what they learn in week one.

Week	Focus	Key Outputs
Week 1 Jun 15-21	Discovery and Setup	Download and explore PDI dataset. Understand field structure and the event/unit relationship. Set up development environment and version control. Agree on tech stack and assign roles. Begin field mapping design.
Week 2 Jun 22-28	Core Pipeline	CSV ingestion and field mapping working end-to-end. First synthetic data generation run (even if rough). Basic validation metrics running. Stats intern begins literature review on validation methodology.
Week 3 Jun 29-Jul 5	Validation and UI	Executive dashboard drafts. Researcher validation report drafts. UI refinement. Stats intern proposes final metric approach and documents reasoning.
Week 4 Jul 6-10	Polish and Package	Application packaged as Windows .exe. All outputs reviewed for clarity and correctness. Final validation report complete. Demo-ready application.

12. Risks and Mitigations

Risk	Likelihood	Mitigation
PyInstaller packaging of Streamlit app requires non-trivial configuration	High	Plan at least two full days for packaging. Research known compatibility issues early. Consider Gradio as an alternative if Streamlit packaging is intractable.
SDV struggles with the event/unit relational structure	Medium	Review SDV's HMA (Hierarchical Modeling Algorithm) documentation in week one. Have a fallback plan to synthesize event-level and unit-level data separately.
Public PDI dataset has inconsistent or missing fields	Medium	Download two or three datasets in week one and assess field coverage. Use the one with the most complete field structure for development.
Stats intern and dev intern disagree on metric approach	Low	Escalate to David Schwindt. The PRD prescribes starting targets; the stats intern can propose changes but final decisions go through the point of contact.
Scope creep into Phase 2 features	Medium	Refer back to Section 5.2. If someone proposes a feature not in Section 5.1, it is Phase 2 by definition. Log it and keep moving.

13. Point of Contact

David Schwindt is the point of contact for this project. Reach out with questions about project direction, scope decisions, data governance, or anything not covered in this document. Mentors through the Hack Your Summer program can help with technical unblocking during office hours, but decisions about project direction go through David.

When in doubt, build what is in scope, document what you are unsure about, and ask.